

Probabilistic Graphical Models & Probabilistic AI

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Lecture 15: A Linear Intro to Generative Models

October 27, 2025

Reading: See course homepage



Exam Results





Course Schedule / Calendar

Week	Lecture Dates	Topic	Assignments
Module 1: Introduction and Foundations			
1	9/3	Course Introduction	
2	9/8, 9/10	A Brief History of DL, Statistics / linear algebra / calculus review	HW1
3	9/15, 9/17	Single-layer networks Parameter Optimization and Gradient Descent	
4	9/22, 9/24	Automatic differentiation with PyTorch, Cluster and cloud computing resources	HW 2
Module 2: Neural Networks			
5	9/29, 10/1	Multinomial logistic regression, Multi-layer perceptrons and backpropagation	
6	10/6, 10/8	Regularization Normalization / Initialization	HW 3
7	10/13, 10/15	Optimization, Learning Rates CNNs	Project Proposal
8	10/20, 10/22	Review, Midterm Exam	In-class Exam

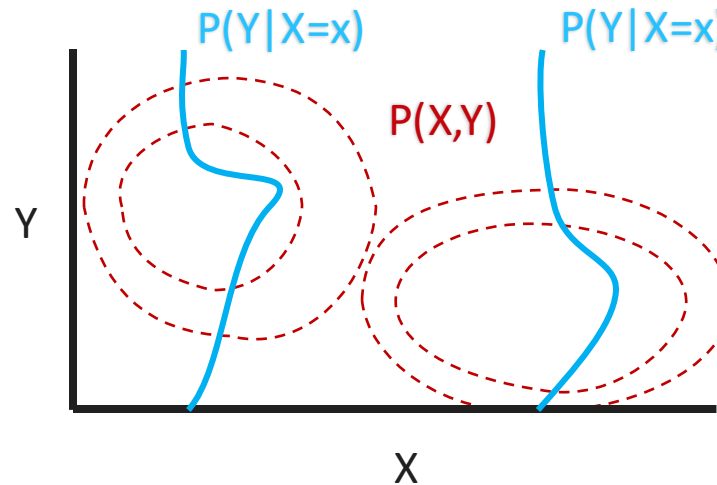
Week	Lecture Dates	Topic	Assignments
Module 3: Intro to Generative Models			
9	10/27, 10/29	A Linear Intro to Generative Models, Factor Analysis, Autoencoders, VAEs	
10	11/3, 11/5	Generative Adversarial Networks, Diffusion Models	Project Midway Report
Module 4: Large Language Models			
11	11/10, 11/12	Sequence Learning with RNNs Attention, Transformers	HW4
12	11/17, 11/19	GPT Architectures, Unsupervised Training of LLMs	
13	11/24, 11/26	Supervised Fine-tuning of LLMs, Prompts and In-context learning	HW5
14	12/1, 12/3	Foundation models, alignment, explainability Open directions in LLM research	
15	12/8, 12/10	Project Presentations	Project Final Report
16	12/17	Final Exam	Final Exam



A Linear Intro to Generative Models

Generative and Discriminative Models

- **Generative:**
 - Models the joint distribution $P(X, Y)$.
- **Discriminative:**
 - Models the conditional distribution $P(Y|X)$.



Two paths to $P(Y|X)$

- **Discriminative:**

Observe X, Y



Learn $P(Y|X)$

- **Generative:**

- Learn $P(X|Y), P(Y)$
- Calculate $P(X) = \int_Y P(X, Y) dY$

Observe X, Y



$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Two paths to classification

- **Discriminative:**

Observe X, Y



$$\hat{Y} = \operatorname{argmax}_Y P(Y|X)$$

- **Generative:**

- Learn $P(X|Y), P(Y)$
- Calculate $P(X) = \int_Y P(X, Y) dY$



Observe X, Y

$$\hat{Y} = \operatorname{argmax}_Y P(X|Y)P(Y)$$

Example Discriminative Model: Logistic Regression

Observe X, Y



Learn $P(Y|X)$

- Parameterize:

- $P(Y = 1|X) = \sigma(\theta^T X)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.
- $P(Y = 0|X) = 1 - P(Y = 1|X)$
- Why this parameterization?**

$$\begin{aligned} \log \frac{P(Y = 1|X)}{P(Y = 0|X)} &= \log \frac{\sigma(\theta^T X)}{1 - \sigma(\theta^T X)} \\ &= \log \frac{\frac{1}{1+e^{-\theta^T X}}}{1 - \frac{1}{1+e^{-\theta^T X}}} = \log \frac{\frac{1}{1+e^{-\theta^T X}}}{\frac{(1+e^{-\theta^T X}) - 1}{1+e^{-\theta^T X}}} = \log \frac{\frac{1}{1+e^{-\theta^T X}}}{\frac{e^{-\theta^T X}}{1+e^{-\theta^T X}}} \\ &= \log \frac{1}{e^{-\theta^T X}} = \log e^{\theta^T X} = \theta^T X \end{aligned}$$

Example Discriminative Model: Logistic Regression

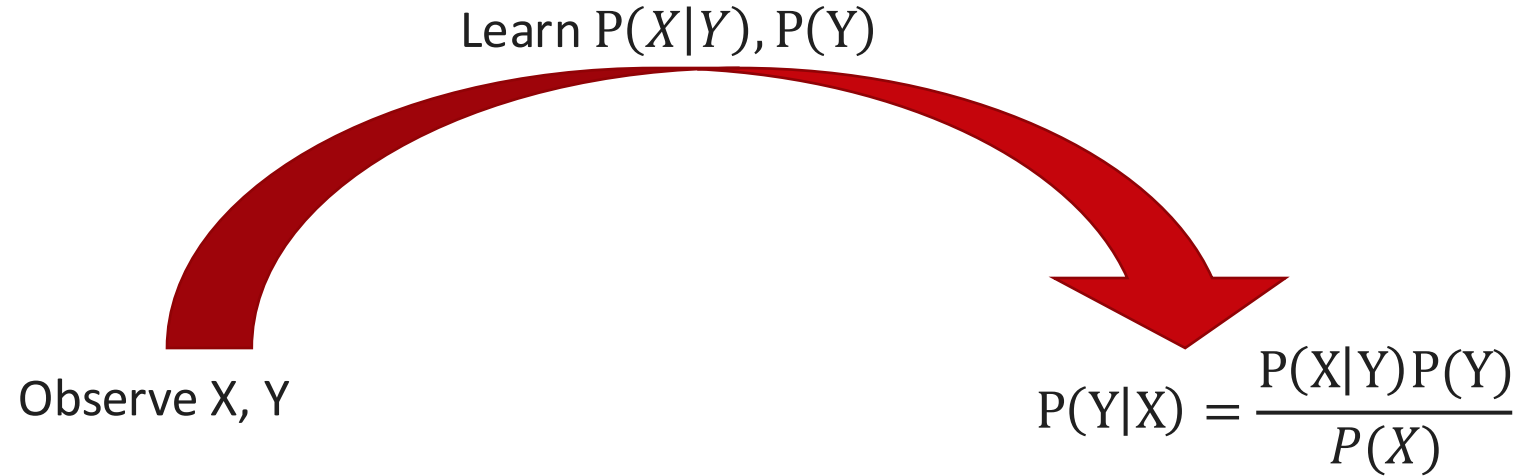
Observe X, Y



Learn $P(Y|X)$

- Parameterize:
 - $P(Y = 1|X) = \sigma(\theta^T X)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.
 - $P(Y = 0|X) = 1 - P(Y = 1|X)$
- Estimate $\hat{\theta}$ from observations:
 - $\hat{\theta} = \operatorname{argmax}_{\theta} \prod_i P(Y_i|X_i; \theta)$
 $= \operatorname{argmax}_{\theta} \sum_i [Y_i \log \sigma(\theta^T X_i) + (1 - Y_i) \log(1 - \sigma(\theta^T X_i))]$
- Calculate $P(Y = 1|X) = \sigma(\theta^T X)$

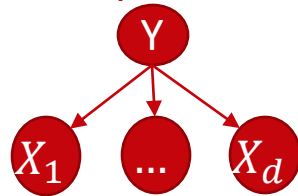
Example Generative Model: Naïve Bayes



- Parameterize:

- Assume $P(X|Y) = \prod_{j=1}^d P(X_j|Y)$,
- $P(X_j|Y) = N(\mu_{jk}, \sigma_{jk}^2)$

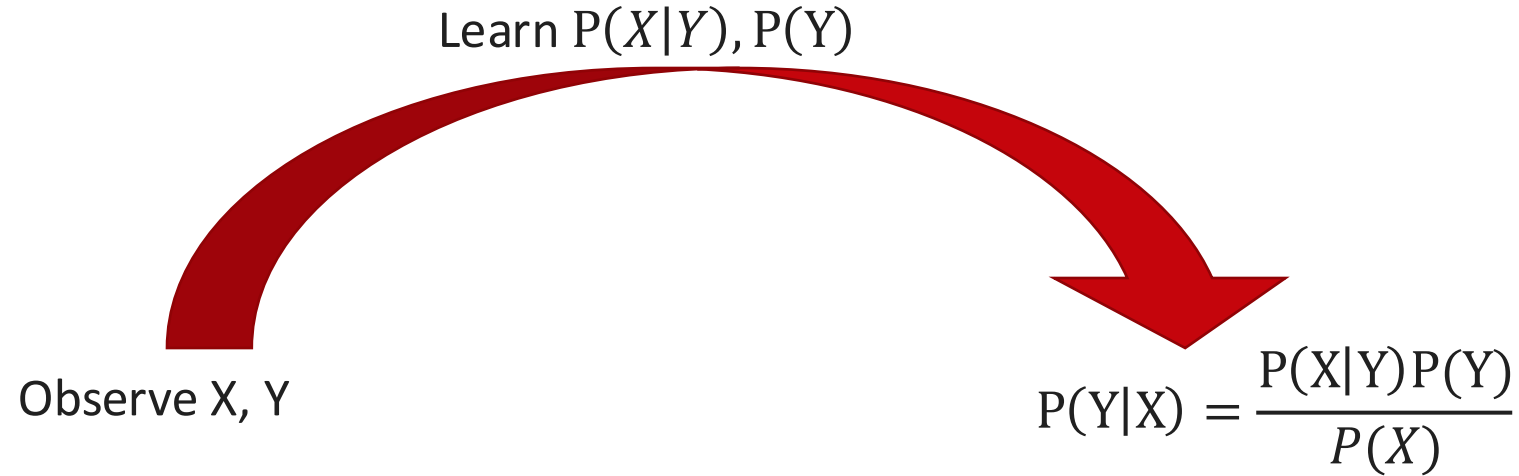
Conditional independences of features $X \mid Y$



$$P(Y = k) = \frac{\text{\# of samples with } Y=k}{\text{Total samples}}$$

Frequency of labels

Example Generative Model: Naïve Bayes



- Parameterize:

- Assume $P(X|Y) = \prod_{j=1}^d P(X_j|Y)$,

$$P(Y = k) = \frac{\text{\# of samples with } Y=k}{\text{Total samples}}$$

- Estimate:

- $\hat{\mu}, \hat{\sigma} = \operatorname{argmax}_{\mu, \sigma} P(X|Y)$

- Calculate $P(Y = 1|X) = \frac{\prod_{j=1}^d P(X_j|Y = 1)P(Y=1)}{P(X)}$

Summary

- **Discriminative:**

Observe X, Y



Learn $P(Y|X)$

- **Generative:**

- Learn $P(X|Y), P(Y)$
- Calculate $P(X) = \int_Y P(X, Y) dY$

Observe X, Y



$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

What about MAP / Regularization?

Logistic Regression:

Observe X, Y



Learn $P(Y|X; \theta)$

Prior $P(\theta)$



- Parameterize:
 - $P(Y = 1|X) = \sigma(\theta^T X)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.
 - $P(Y = 0|X) = 1 - P(Y = 1|X)$
- Estimate $\hat{\theta}$ from observations:
 - $\hat{\theta} = \operatorname{argmax}_{\theta} \prod_i P(Y_i|X_i; \theta) P(\theta)$
 $= \operatorname{argmax}_{\theta} \sum_i [Y_i \log \sigma(\theta^T X_i) + (1 - Y_i) \log(1 - \sigma(\theta^T X_i))] - R(\theta)$
- Calculate $P(Y|X)$

Discriminative vs Generative Models

- Discriminative models optimize the conditional likelihood:

$$\widehat{\theta}_{disc} = \operatorname{argmax}_{\theta} P(Y|X; \theta)$$

- Generative models optimize the joint likelihood:

$$\widehat{\theta}_{gen} = \operatorname{argmax}_{\theta} P(X, Y; \theta)$$

Are these the same optimization?

Discriminative vs Generative Models

- Discriminative models optimize the conditional likelihood:

$$\widehat{\theta}_{disc} = \operatorname{argmax}_{\theta} P(Y|X; \theta) = \operatorname{argmax}_{\theta} \frac{P(X|Y; \theta)P(Y; \theta)}{P(X; \theta)}$$

- Generative models optimize the joint likelihood:

$$\widehat{\theta}_{gen} = \operatorname{argmax}_{\theta} P(X, Y; \theta) = \operatorname{argmax}_{\theta} P(X|Y; \theta)P(Y; \theta)$$

Are these the same optimization?

Same optimization when $P(X; \theta)$ is invariant to θ

Logistic Regression vs Naïve Bayes

Logistic Regression	Naïve Bayes
Discriminative	Generative
Defines $P(Y X; \theta)$	Defines $P(X, Y; \theta)$
Estimates $\widehat{\theta}_{lr} = \operatorname{argmax}_{\theta} P(Y X; \theta)$	Estimates $\widehat{\theta}_{nb} = \operatorname{argmax}_{\theta} P(X, Y, \theta)$
Lower asymptotic error on classification	Higher asymptotic error on classification
Slower convergence in terms of samples	Faster convergence in terms of samples

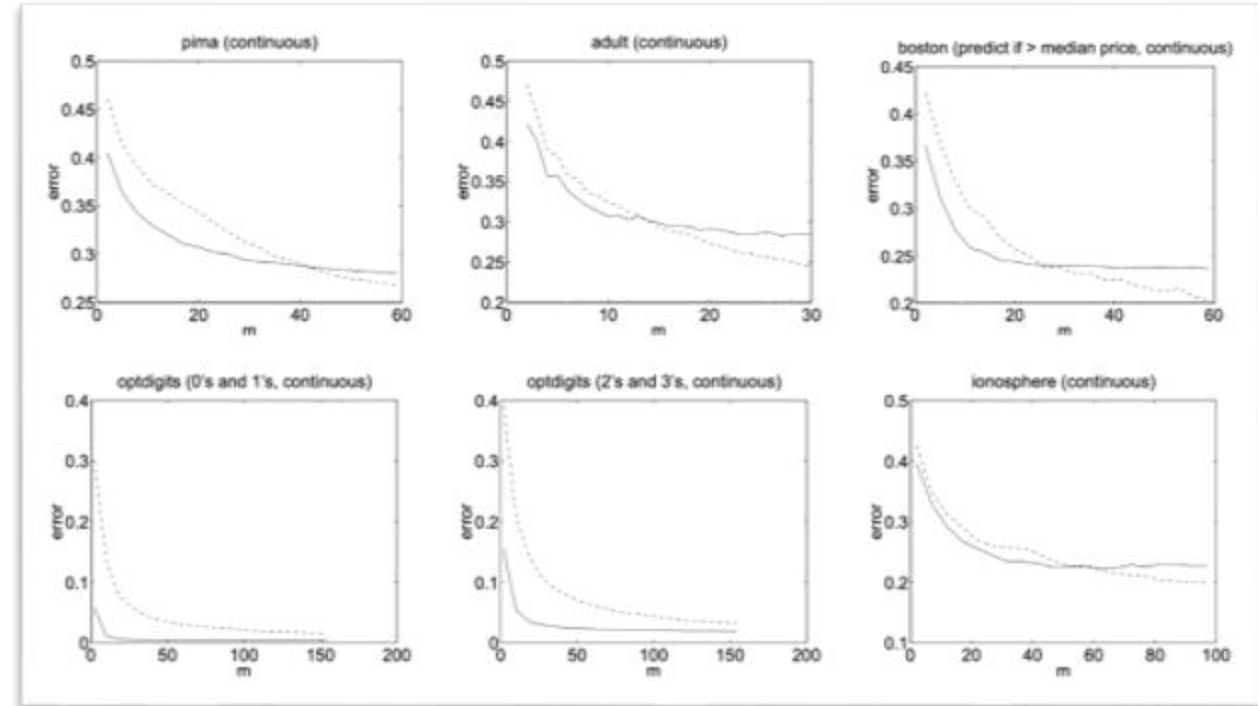
Andrew Ng's Insight

- “While discriminative learning has lower asymptotic error, a generative classifier may also approach its (higher) asymptotic error much faster.”

Why?

..... LR
 — NB

Ng & Jordan 2001



Andrew Ng's Insight

- “While discriminative learning has lower asymptotic error, a generative classifier may also approach its (higher) asymptotic error much faster.”
- Underlying assumption of this statement:
 - Generative models of the form $P(X, Y, \theta)$ make **more simplifying assumptions** than do discriminative models of the form $P(Y|X, \theta)$.
 - **Not always true**
 - “So far there is no theoretically correct, general criterion for choosing between the discriminative and the generative approaches to classification of an observation $\mathbf{x}$ into a class y ; the choice depends on the relative confidence we have in the correctness of the specification of either $p(y|\mathbf{x})$ or $p(\mathbf{x}, y)$ for the data.”

[Xue & Tittering 2008](#)

Modern Deep Generative Models

January 27, 2025:



Questions?

